

Enhancing Safety in Collaborative Cable-Driven Parallel Robots: Contact Distinction and Management for Carrying Tasks

Hanbang Gao, Christine Chevallereau, and Stéphane Caro

Abstract—Cable-Driven Parallel Robots (CDPRs) have shown significant potential in industrial applications due to their large workspace, high payload capacity, and flexibility. Nonetheless, ensuring safe and efficient human-robot collaboration, particularly during carrying tasks, remains an open problem. This paper primarily contributes a cable tension-based metric for detecting contact events, combined with a frequency-domain analysis of estimated external wrenches to distinguish different contact scenarios. Building upon this framework, three common contact scenarios are then addressed: (i) payload placement and removal without rigid attachment to the moving platform (SC1), (ii) unintentional human-cable collisions (SC2), and (iii) human-platform collisions (SC3). Specific management strategies are then proposed, including real-time mass estimation for payload handling, cable tension release for human-cable collisions, and a compliant trajectory controller for human-platform collisions. Experimental validations on a CDPR prototype demonstrate accurate scenario classification and safe contact handling without compromising overall productivity. By facilitating safer interactions, improved adaptability, and reliable handling of diverse contact events, this work expands the applicability of collaborative CDPRs in real-world industrial and logistic settings.

Note to Practitioners—In many industrial applications, robots operating alongside human workers must quickly detect and respond to contact events to maintain safety and efficiency. Traditional methods often rely on force thresholds, which can introduce delays or false alarms, especially in dynamic tasks where rapid collisions must be addressed. Our work proposes an alternative approach that utilizes a frequency-domain analysis of sensed cable tensions and a weight-score system to robustly distinguish among various contact scenarios. By examining tension signals in the frequency domain, more nuanced information can be extracted, enabling faster and more accurate responses than threshold-based methods. From a practical standpoint, this technique is relatively straightforward to implement with various sensors, and it can be adapted to robots composed of different materials and link configurations by tuning the weight-score system for each new setup. Its primary benefit is improved contact distinction with high precision, which can enhance operator safety and reduce downtime without requiring extensive hardware modifications. However, like any sensor-driven control method, performance depends on adequate calibration and consistent signal quality. Future research might focus on integrating additional sensing modalities (e.g., vision or proximity sensors) and exploring broader deployment in diverse collaborative robot systems.

Index Terms—Tendon/Wire Mechanism, Parallel Robots, Human-Robot Collaboration, Physical Human-Robot Interaction, Safety in HRI.

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I. INTRODUCTION

PARALLEL robots have attracted significant attention for their high precision, rigidity, and efficiency, achieved by employing multiple actuators arranged in parallel for stable and versatile motion [1]. Among these, Cable-Driven Parallel Robots (CDPRs) stand out due to their lightweight structure and relatively large workspaces, enabled by reconfigurable cable arrangements [2], [3].

CDPRs feature a high payload-to-weight ratio, large workspace, cost efficiency, and high reconfigurability [4], making them particularly useful for industrial applications such as precision manipulation tasks (e.g., machining, painting, medical procedures) [5]–[7], as well as material handling, logistics, construction, and assembly [8]. In summary, the combination of a large reachable workspace, flexible reconfiguration, and relatively simple hardware renders CDPRs ideal for tasks that span from delicate manipulation to heavy-load transport. Although collaborative CDPRs have emerged as a novel research topic to meet modern demands for flexibility and performance, only limited studies have been conducted in this area.

In the context of Industry 4.0, collaborative robots are critical for harnessing human-robot synergy, optimizing workflows, and meeting increasing demands for flexibility and customization [9]–[11]. Collaborative CDPRs enhance system versatility and adaptability by executing tasks that involve direct human interaction, such as rehabilitation, human augmentation, and co-manipulation in industrial settings. Unlike conventional CDPRs, which are designed for predefined tasks, collaborative CDPRs can adjust to dynamic and unpredictable environments. They are often equipped with sensors and algorithms for collision detection and avoidance, enabling safe operation around humans without the need for extensive physical barriers. Despite these benefits, collaborative CDPRs with physical Human-Robot Interaction (pHRI) introduce new challenges in intentional human-robot communication and unintentional collision management within the shared workspace. Figure 1 depicts a representative scenario in which a human operator is assisting the CDPR in carrying a payload, highlighting the integration of human effort with robotic assistance in an industrial or logistic context.

Human-robot collaboration has been widely studied in serial and parallel robots [12], [13], where compliance-based control strategies (e.g., impedance and admittance) relying on force sensing are regarded as effective solutions [14]. Recent work

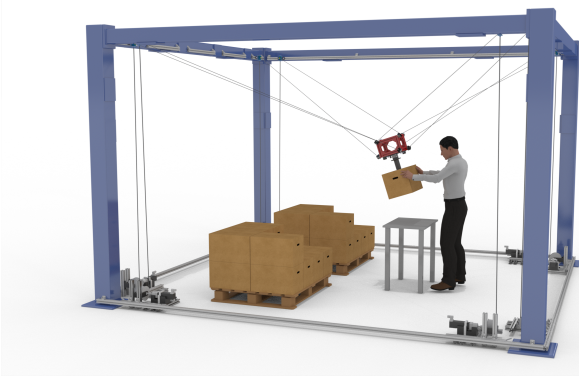


Fig. 1: Human-robot collaborative setup in which a human operator is loading a payload onto a CDPR

has extended these methods to collaborative CDPRs, examining, for instance, how CDPR stiffness affects transparency (the robot's ability to move in the user's intended direction) during co-manipulation [15] and proposing advanced control strategies for enhanced physical interaction [16]. Such approaches can enable co-manipulation without predefined trajectories via direct force sensor measurements. However, human operators in CDPR-based carrying tasks may interact with cables or the Moving Platform (MP), not necessarily with an instrumented handle that embeds a force sensor.

Ensuring safe interaction in a CDPR that follows a predefined trajectory demands sophisticated real-time collision avoidance, detection, and management. Thin cables can exert significant pressure when colliding with a human [17], and the MP can reach high speed and acceleration, posing potential risks to both operator safety and system stability. Conventional CDPRs also encounter self-interference issues involving cable-cable or cable-MP interference [18], [19], along with failures in cables, motors, or encoders [20], [21]. Some studies address known obstacles by generating collision-free trajectories [22] or reconfiguring the CDPR [23], [24] for collision avoidance, while others consider geometric and dynamic effects such as cable wrapping around cylindrical surfaces [25], [26]. In unknown and human-involved environments, collision management has mainly focused on human-cable contacts, where tension measurements detect collisions and prompt cables to be released to a safer tension without altering the original trajectory [17], [27], [28]. Nevertheless, existing research has not addressed human-MP collisions with concrete experimental results, likely due to the lack of direct contact-force measurement and associated safety concerns.

Integrating safety management into collaborative CDPR frameworks remains a cutting-edge direction for industrial implementation. A general-purpose industrial motion-control architecture proposed in [29] describes a two-layer structure containing a safety controller and a motion controller, applicable to toolpath scenarios. Another approach addresses collaborative manipulation of a cable-suspended platform activated by a serial robot [30], distinguishing collisions from cooperative contacts via frequency analysis. However, applying these strategies to collaborative CDPRs introduces further

complexity. Existing contact distinction methods often do not differentiate human-cable from human-MP collisions, despite their distinct dynamics [30], [31], and pure admittance control without predefined trajectories rarely captures common industrial requirements for higher robot stiffness [15], [16], [32]. Industrial scenarios typically involve autonomous carrying tasks with intermittent human participation, highlighting the need for real-time intent detection and collision management to support seamless collaboration.

To advance research in collaborative CDPRs, this work proposes a theoretical and experimental framework for collaborative CDPR carrying tasks. The framework classifies contact events into three main scenarios: (i) placement/removal of payloads, (ii) human-cable collision, and (iii) human-MP collision. Each scenario is addressed with dedicated strategies to ensure operator safety and system robustness. These strategies are experimentally validated with accompanying video demonstrations.

The objectives and novel contributions of this work include:

- Proposing a framework to detect and distinguish different types of contact in CDPRs (human-cable collision vs. human-MP collision vs. payload placement/removal).
- Developing appropriate management strategies for each contact scenario (to ensure safety, stability, and efficient collaboration).
- Validating the approach through experiments that demonstrate how the system handles contact events in real time while preserving operator safety and overall productivity.

To the authors' knowledge, no existing work explicitly considers collisions with multiple robot components, including both cables and the MP, while providing experimental validation for handling these diverse scenarios. The proposed contact distinction approach fills this gap by enabling the robot to maintain its planned trajectory and adapt in real time as soon as human intent or unforeseen collisions are detected, thereby addressing the practical demands of industrial environments.

The remainder of this article is organized as follows. Section II describes the three primary contact scenarios in human-assisted carrying tasks and introduces the proposed framework for addressing each scenario, along with the experimental setup. Section III details the robot kinematic modeling, control scheme, and trajectory planning. Section IV presents the approach for contact detection. Section V outlines the frequency analysis-based methodology used to distinguish among the three scenarios. Section VI discusses specific management strategies tailored to each scenario and introduces a method for determining the termination of each contact event. Finally, Section VII concludes the paper with a summary of the key findings and avenues for future work.

II. METHODOLOGICAL FRAMEWORK

In this section, the three common contact scenarios of human-assisted CDPR carrying tasks in industrial or logistics settings are introduced to define the scope of this study. Relevant assumptions are then presented. The following subsection explains the proposed management framework for these scenarios through an integrated flowchart. Lastly,

the PLATFORMNAME platform and associated experimental setup are described, showing how the proposed methods are implemented.

A. Contact Scenarios in Carrying Tasks

CDPRs in industrial and logistics domains offer a large workspace and a favorable payload-to-mass ratio, making them suitable for cargo transportation. They can be integrated into flexible manufacturing systems or automated warehouses for various handling operations. Figure 1 illustrates a typical scenario: a human operator placing a box onto the MP of a CDPR. In more general industrial processes, the MP may continue following a reference trajectory while payloads are added or removed. This is referred to as Scenario Case 1 (SC1), indicating that placement and removal can occur even when the MP is not in static equilibrium.

The shared workspace between the robot and human can lead to unintentional collisions involving the operator and either the cables or the MP during operation. Human–cable contact and human–MP contact constitute Scenario Cases 2 (SC2) and 3 (SC3), respectively. These contacts are treated as unintentional collisions rather than cooperative manipulation. Minor interactions are not detected (as explained in Section IV), and once a collision is formally detected, it is managed to ensure operator safety and CDPR stability.

- **SC1: Payload Placement and Removal.** The payload is placed on top of the MP without being rigidly attached to the MP frame. It is assumed that the operator places the payload gently (with a minimum mass of approximately 500 g and a maximum of 10 kg). During payload placement, there is no direct human–MP contact, and the payload is not expected to slide on the platform. There are no further assumptions regarding payload removal beyond the absence of direct human–MP contact.
- **SC2: Human–Cable Collision.** It is assumed that only one cable is involved in a collision at any given time, as illustrated in Fig. 2. Once the collision is removed and properly managed, a subsequent human–cable collision may occur. The collided cable is assumed to lie within a Wrench Feasible Workspace (WFW) that allows tension release, effectively reducing the cable tension to zero.

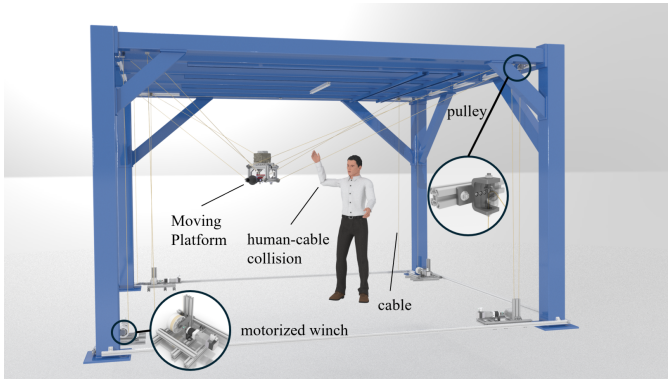


Fig. 2: A human-cable collision and the PLATFORMNAME experimental setup

- **SC3: Human–MP Collision.** This scenario considers contact between the human operator and the MP frame, typically from the side of the platform. Since the MP transports payloads primarily in a horizontal plane, a direct vertical (top-to-bottom or bottom-to-top) collision is less likely. Collisions are assumed to be instantaneous rather than gradual force applications over time, which still represent realistic industrial cases.
- **Non-Concurrent Contacts.** All three scenarios are assumed to occur sequentially rather than simultaneously. A new contact scenario is initiated only after the currently managed contact has been fully resolved. Note that payload placement and payload removal are regarded as two separate events, and both human–cable and human–MP collisions may occur after a payload has been placed.

B. Proposed Framework

Figure 3 depicts the proposed workflow. After system initialization, an estimation of the payload mass is performed to update the MP's total load for inverse modeling in Cartesian-to-motor space. The CDPR executes the reference trajectory based on this dynamic information. If contact is detected via real-time monitoring using the cable tension sensors, a contact distinction procedure is initiated to determine whether SC1, SC2, or SC3 has occurred.:

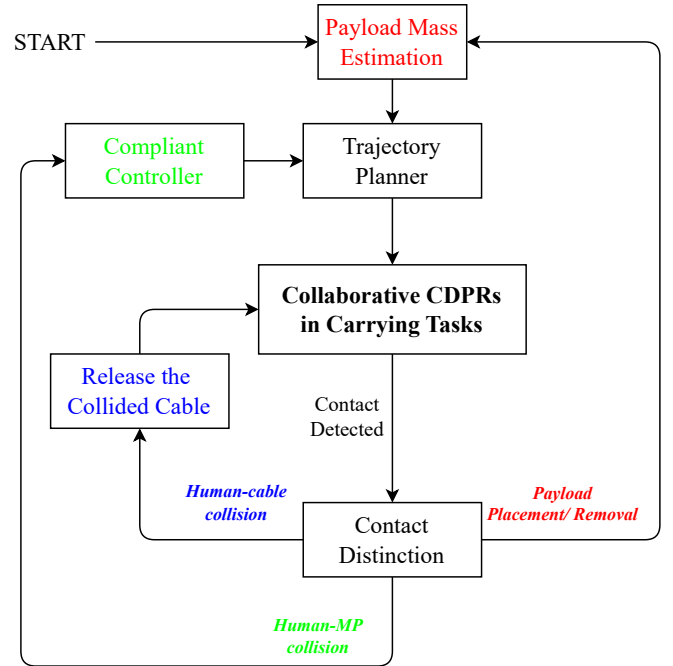


Fig. 3: Workflow of the proposed system for contact detection and management. Each box represents a task phase, while the colored text and corresponding boxes highlight the scenario detected after the contact distinction. Red corresponds to SC1, blue to SC2, and green to SC3, with each color denoting the scenario management strategy.

- **SC1:** When a payload is placed or removed, its mass is re-estimated using the CDPR dynamic model. This

estimation is performed over a period of time until the changes become negligible, and then the mass is fixed. This procedure is necessary to accurately detect and distinguish human-robot contacts.

- **SC2:** In the event of a human–cable collision, tension in the collided cable is released to a permissible level while the reference trajectory continues.
- **SC3:** For a human–MP collision, the reference trajectory is modified using a compliant controller sub-modules. The compliance sub-module interprets external forces to adjust the reference trajectory, and ensures a return to the original trajectory once the collision is resolved. The scenario ends when the modified and original trajectories converge below a predefined threshold.

C. Experimental Setup

The PLATFORMNAME platform at LABNAME is a CDPR prototype featuring eight suspended cables and dedicated actuation units, as shown in Fig. 2. Each actuation chain comprises a motor with an integrated gearhead, a winch, and a cable attached to the MP. The system is managed by a supervisory PC that communicates with a real-time computer (dSPACE), which regulates the torque commands for all eight motors.

Two types of sensor data are available at a sampling rate of 1 kHz:

- 1) **Motor Encoders:** Provide joint angles $\mathbf{q}_m$ to the real-time PC.
- 2) **Dynamometers:** Positioned near cable attachment points, transmitting cable tension measurements τ_m to the real-time PC.

An embedded exponential moving average (EMA) filter processes τ_m to yield the filtered tension τ_{mf} [28]. The overall architecture facilitates closed-loop control and real-time monitoring, enabling the implementation of the proposed framework for contact detection, distinction, and management.

III. ROBOT MODELING, CONTROL, AND TRAJECTORY PLANNING

In this section, the modeling of the CDPR, the control scheme corresponding to Fig. 3, and the trajectories used in the experiments are presented. The methodology includes the Inverse Geometric Elasto-static Model (IGEM), the Inverse Kinematic Model (IKM), and the dynamic model.

A. Robot Modelling

1) **IGEM:** The IGEM establishes a mapping between the Cartesian pose of the MP and the corresponding motor angles in the joint space. It incorporates pulley geometry [16] and linear cable elasticity [33], thereby ensuring improved accuracy in cable length computation by explicitly accounting for the dependence on the desired cable tension τ_d . Although more sophisticated nonlinear elasticity models have been studied [34], this work assumes linear elasticity behavior within the specified load range. The axial stiffness (ES) of the cable is identified through dedicated experiments rather than relying on the manufacturer's values. A light 0.7 mm-diameter cable

with a mass of 4 g/m is employed, and as sagging effects are negligible for payloads exceeding 5 kg, the cable is considered massless for this study. An accurate IGEM is vital for precise Cartesian trajectory tracking of the MP.

2) **IKM:** For CDPRs, the forward Jacobian matrix is equals to $-\mathbf{W}^\top$, which transforms the platform velocity in Cartesian coordinates, $\mathbf{v}$, into the cable unwinding velocities $\dot{\mathbf{l}}$:

$$-\mathbf{W}^\top \mathbf{v} = \dot{\mathbf{l}}, \quad \text{with } \mathbf{v} = \begin{bmatrix} \dot{\mathbf{p}} \\ \dot{\boldsymbol{\omega}} \end{bmatrix}, \quad (1)$$

where $\mathbf{p}$ and $\dot{\mathbf{p}}$ is the Cartesian position and velocity of the MP, $\boldsymbol{\omega}$ is the angular velocity of the MP expressed in the base frame, and $\mathbf{l}$ is the vector of cable lengths from anchor positions to pulley exit points. The matrix $\mathbf{W}$, referred to as the wrench matrix.

3) **Dynamic Modeling:** Following [35], the dynamic model of a CDPR is

$$\mathbf{W}\boldsymbol{\tau} - \mathbb{I}_p \dot{\mathbf{v}} - \mathbf{C}\mathbf{v} + \mathbf{w}_e + \mathbf{w}_g = 0, \quad (2)$$

where $\boldsymbol{\tau}$ is the vector of cable tensions, $\mathbb{I}_p$ is the spatial inertia of the platform, and $\mathbf{C}$ is the matrix of centrifugal and Coriolis wrenches. The term $\mathbf{w}_e$ represents external wrenches (e.g., collisions and contacts), and $\mathbf{w}_g$ is the gravitational wrench. The gravity wrench $\mathbf{w}_g$ due to gravity vector $\mathbf{g} = [0, 0, -g]^\top$ is

$$\mathbf{w}_g = \begin{bmatrix} m\mathbf{I}_3 \\ \hat{\mathbf{M}}\mathbf{S}_p \end{bmatrix} \mathbf{g}, \quad (3)$$

where m is the platform mass, $g = 9.81 \text{ m s}^{-2}$, and $\mathbf{I}_3$ denotes the 3×3 identity matrix in $\mathbb{R}^3$. The first momentum of the moving platform with respect to the MP frame is represented by $\mathbf{M}\mathbf{S}_p$, and its corresponding skew-symmetric matrix is denoted by $\hat{\mathbf{M}}\mathbf{S}_p$. Note that the geometric center and the center of gravity of the MP do not coincide, resulting in nonzero first moments. The spatial inertia $\mathbb{I}_p$ is defined as

$$\mathbb{I}_p = \begin{bmatrix} m\mathbf{I}_3 & -\hat{\mathbf{M}}\mathbf{S}_p \\ \hat{\mathbf{M}}\mathbf{S}_p & \mathbf{I}_p \end{bmatrix}, \quad (4)$$

where the off-diagonal terms account for the first moments arising from the discrepancy between the geometric center and the center of gravity, and $\mathbf{I}_p$ is the inertia tensor of the platform. It can be derived from the inertia tensor $\mathbf{I}_g$ in the MP frame using the Huygens–Steiner theorem:

$$\mathbf{I}_p = \mathbf{R}_p \mathbf{I}_g \mathbf{R}_p^\top - \frac{\hat{\mathbf{M}}\mathbf{S}_p \hat{\mathbf{M}}\mathbf{S}_p}{m}. \quad (5)$$

In (2), the matrix $\mathbf{C}$ yields centrifugal and Coriolis wrenches:

$$\mathbf{C}\mathbf{v} = \begin{bmatrix} \hat{\boldsymbol{\omega}} \hat{\boldsymbol{\omega}} \mathbf{M}\mathbf{S}_p \\ \hat{\boldsymbol{\omega}} \mathbf{I}_p \boldsymbol{\omega} \end{bmatrix}, \quad (6)$$

where $\hat{\boldsymbol{\omega}}$ is the skew-symmetric matrix of $\boldsymbol{\omega}$. This formulation underpins the control design and analysis for human-assisted operations.

B. Control Architecture

Figure 4 summarizes the proposed control scheme for the PLATFORMNAME platform. Three main parts, highlighted

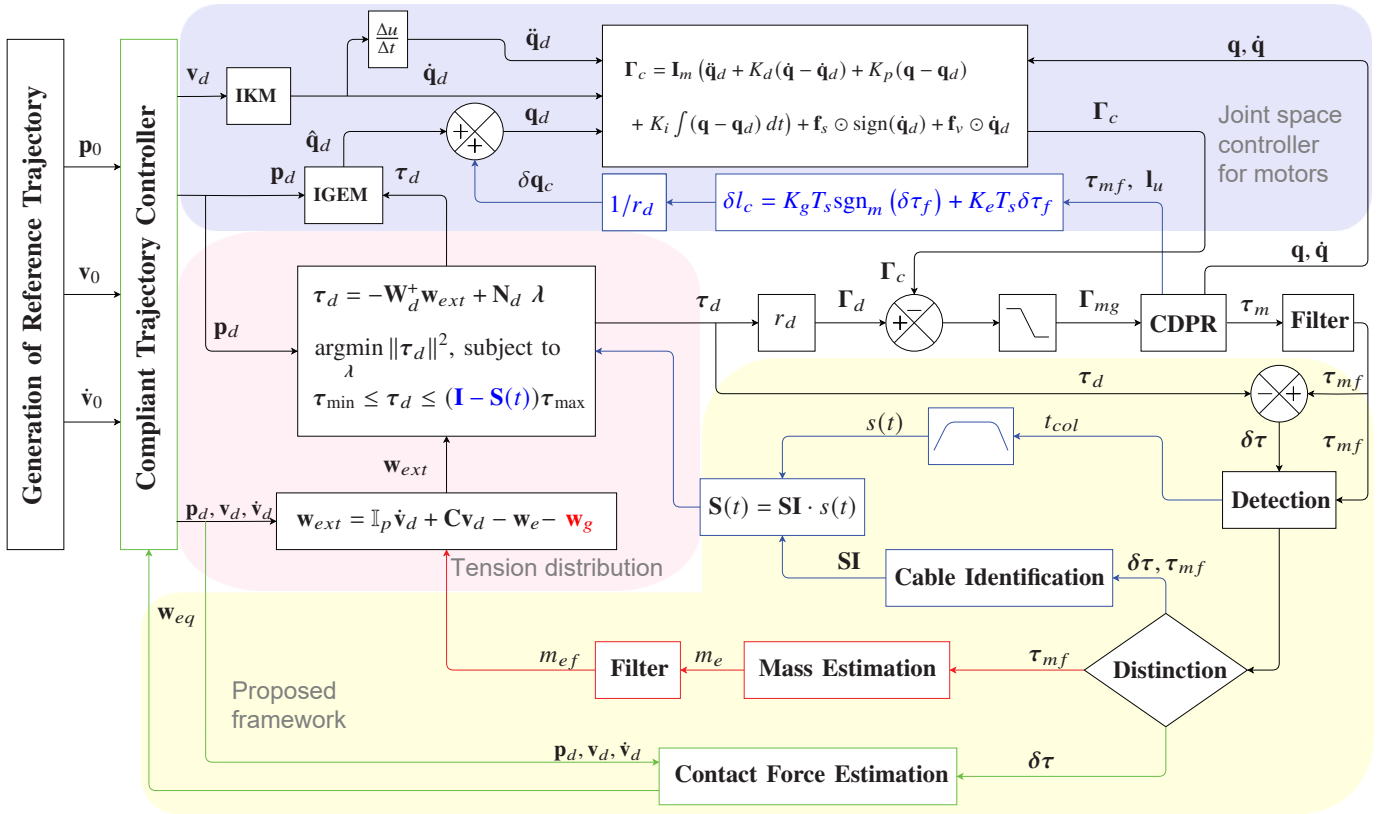


Fig. 4: Proposed control scheme of the PLATFORMNAME platform. Black blocks indicate the standard controller for nominal robot operations. Each scenario case (SC1 in red, SC2 in blue, and SC3 in green) is managed by a distinct strategy following contact detection and is activated only when the corresponding event is distinguished.

in colored patches, constitute the architecture:

- **Purple (Joint Space Controller):** Implements a computed torque control law with PID correction, driving motors from measured angles $\mathbf{q}$ to desired angles $\mathbf{q}_d$. Desired values $\mathbf{q}_d$ are computed via the IGEM, enabling accurate tracking in joint space.
- **Pink (Tension Space Controller):** Translates desired motion commands of the MP into desired cable tensions τ_d . A feed-forward term $\mathbf{w}_{ext}$ derived from the inverse dynamic model compensates for inertia, Coriolis, and gravitational effects.
- **Yellow (Proposed System for Contact Distinction and Management):** Monitors sensor signals in real time, identifies collisions, and applies the appropriate strategy (SC1, SC2, or SC3) based on the contact distinction framework.

Black blocks within Fig. 4 represent components dedicated to nominal operation, including the motor-gearbox torque Γ_{mg} . A saturation function ensures safe operating limits. The joint space controller outputs Γ_c after accounting for frictional effects and desired accelerations, while the tension space controller generates Γ_d to regulate cable tensions. The combination of Γ_c and Γ_d forms the total command Γ_{mg} .

1) *Joint Space Controller:* The desired motor angles $\mathbf{q}_d$ are computed via the IGEM, and a PID corrector with gains K_p , K_i , and K_d forces the tracking error $\mathbf{e} = \mathbf{q}_d - \mathbf{q}$ to converge

to zero:

$$\ddot{\mathbf{e}} + K_d \dot{\mathbf{e}} + K_p \mathbf{e} + K_i \int \mathbf{e} dt = 0. \quad (7)$$

The motor joint acceleration is defined as:

$$\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_d + K_d (\dot{\mathbf{q}} - \dot{\mathbf{q}}_d) + K_p (\mathbf{q} - \mathbf{q}_d) + K_i \int (\mathbf{q} - \mathbf{q}_d) dt. \quad (8)$$

Coulomb and viscous friction are compensated by

$$\Gamma_c = \mathbf{I}_m \ddot{\mathbf{q}} + \mathbf{f}_s \odot \text{sign}(\dot{\mathbf{q}}) + \mathbf{f}_v \odot \dot{\mathbf{q}}, \quad (9)$$

where $\mathbf{I}_m$ is the inertia matrix of the motor-gearbox assembly, and $\odot$ represents element-wise multiplication. $\mathbf{f}_s$ is the static friction and $\mathbf{f}_v$ is the viscous friction.

2) *Tension Space Controller:* To ensure desired motion, the motor torques Γ_d are computed by determining the cable tensions τ_d . A feed-forward term $\mathbf{w}_{ext}$ from the (2) compensates for the system dynamics:

$$\mathbf{w}_{ext} = \mathbb{I}_p \dot{\mathbf{v}}_d + \mathbf{C} \mathbf{v}_d - \mathbf{w}_e - \mathbf{w}_g, \quad (10)$$

where $\mathbf{v}_d$ is the desired platform velocity. The tensions τ_d are optimized using quadratic programming [27]:

$$\min_{\lambda} \|\tau_d\|^2, \quad \text{subject to} \quad \tau_{\min} \leq \tau_d \leq \tau_{\max}, \quad (11)$$

where λ is a two-dimensional vector representing the null space variables. The selection of λ is critical for optimizing the norm of the tensions. The vector $\tau_{\min}$ ensures cable tautness

by maintaining a minimum tension of 1 N per cable, while $\tau_{\max}$ sets the upper limit at 100 N per cable.

The tension vector τ_d is defined as

$$\tau_d = -W_d^+ w_{ext} + N_d \lambda, \quad (12)$$

W_d is the desired wrench matrix derived from the IGEM, $^+$ denotes the Moore–Penrose pseudo-inverse, and N_d is the null space of W_d , satisfying $W_d N_d \lambda = 0_6$. The position of the MP is calculated using the IGEM, which ensures accurate trajectory tracking by accounting for the geometric constraints of the system. The resulting motor–gearbox torque is

$$\Gamma_{mg} = \text{sat}(r_d \tau_d + \Gamma_c), \quad (13)$$

where $\text{sat}(\cdot)$ denotes a saturation function that ensures the output torque of the motors remains within acceptable limits, and r_d represents the winch radius.

C. Trajectory Design

Two trajectories were designed to evaluate system performance:

1) *Static Trajectory*: The MP remains stationary above the test stand with active servo motors. This setup assesses contact detection and distinction under quasi-static conditions.

2) *Dynamic Trajectory*: The MP moves along a 1 m square planar path over 60 s at an average speed of 0.1 m/s. A fifth-order polynomial trajectory governs position, velocity, and acceleration, ensuring smooth transitions and effective contact management during dynamic interactions.

Note that the chosen speed and path length account for the PLATFORMNAME prototype's smaller scale compared to industrial CDPRs (with frames exceeding 20 m). Both length and velocity have been scaled to match the platform's capabilities, avoiding excessive motor loads and cable displacements. Although small-scale CDPRs can achieve higher speeds, this reduced velocity ensures safe and controlled dynamic behavior under the prototype's inherent mechanical constraints, while still demonstrating that the system dynamics extend beyond quasi-static conditions.

IV. CONTACT DETECTION

A. Overview of Contact Detection

A heuristic method for human–cable collision (SC2) detection was introduced in [27] by continuously monitoring cable tension errors. Although originally focused on SC2, this approach can be generalized to detect all three contact scenarios (SC1, SC2, and SC3). The main idea is summarized below.

B. Tension Error Metric

Define the tension tracking error $\delta\tau$ as an eight-dimensional vector representing the difference between the measured and desired cable tensions, as illustrated by the **Detection** block in Fig. 4:

$$\delta\tau = \tau_{mf} - \tau_d, \quad (14)$$

where τ_{mf} is the measured cable tension filtered by an EMA, and τ_d is the desired tension vector. Under normal operation (i.e., when the MP follows its reference trajectory without human interaction), the errors typically remain within 3 N. However, a threshold of 5 N is chosen empirically:

$$T_h = 5 \text{ N}, \quad (15)$$

to balance the following objectives:

- **Collision Sensitivity vs. Safety**: Considering minor contacts (e.g., a light touch between the human and MP) could be safely ignored, any contact producing an error exceeding T_h is managed as a collision event to maintain overall safety.
- **Reduced False Positives**: Allowing a slightly larger error window before declaring contact helps mitigate spurious detections. This approach increases reliability in distinguishing SC1 from SC2 and SC3.

C. Detection Criterion

A contact is declared if any component $\delta\tau_i$ of $\delta\tau$ exceeds the threshold T_h :

$$\exists \delta\tau_i \in \delta\tau : \delta\tau_i > T_h. \quad (16)$$

For SC2, following payload removal, the cable tensions decrease. In contrast, as analyzed in [29], placing an object onto the platform typically involves a fast impact, whereas payload removal is generally a more gradual process. A lower threshold for payload removal is defined as $T_h^- = -4 \text{ N}$. Consequently, a payload removal (SC1) is declared if any component $\delta\tau_i$ of $\delta\tau$ falls below T_h^- :

$$\exists \delta\tau_i \in \delta\tau : \delta\tau_i < T_h^-. \quad (17)$$

D. Experimental Illustration

Figure 5 shows an experiment designed to demonstrate contact detection and distinction in a stationary CDPR. The sequence includes two payload placement events and two payload removal events (SC1), two human–cable collisions (SC2), and two human–MP collisions (SC3). Each scenario is repeated twice to illustrate the generality of the proposed approach.

From the experimental results, three key observations arise:

- **SC1 (Payload Placement/Removal)**: During placement, tension in all cables increases; it decreases when the payload is removed. The placement/removal act introduces small, transient oscillations.
- **SC2 (Human–Cable Collision)**: Some cable tensions (including the collided cable) increase significantly, while others may drop or sag. Due to the relatively low stiffness of the cable, oscillations remain very limited because external forces are largely absorbed.
- **SC3 (Human–MP Collision)**: All cable tensions show more pronounced oscillations, which gradually damp out over time as the controller regulates for the disturbance.

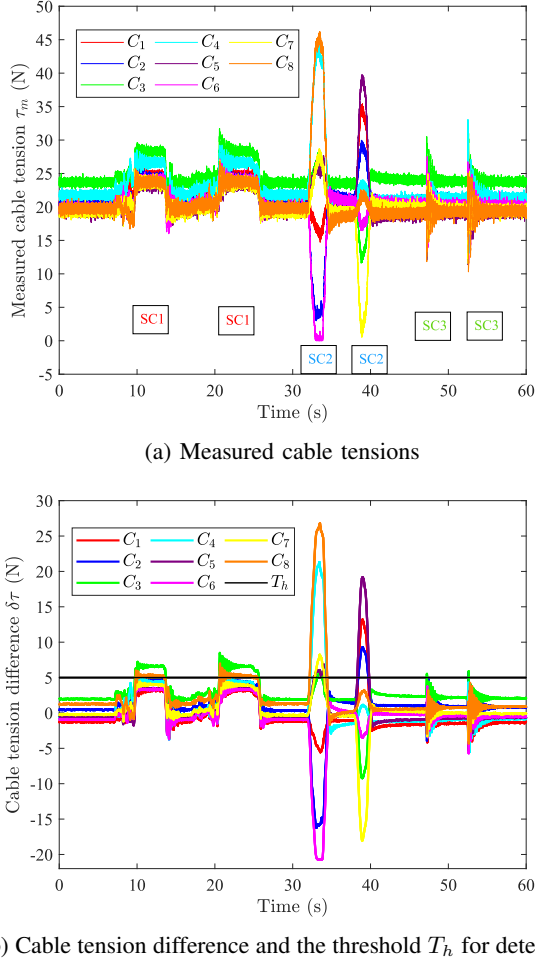


Fig. 5: Measured cable tension τ_m and cable tension difference $\delta\tau$ during a series of contact events with the CDPRS in a stationary configuration. The label C_j denotes the j^{th} cable ($j = 1, 2, \dots, 8$). T_h is the threshold for contact detection.

V. CONTACT DISTINCTION

A. Related Works

Previous investigations on contact distinction have primarily focused on differentiating voluntary from unintentional contacts (collisions). One category of approaches relies on data-driven learning [36], which is computationally intensive and requires extensive data collection. Another category applies heuristic thresholds on contact force or torque [37], [38], although these methods are sensitive to the chosen threshold and often struggle to distinguish contacts with similar force profiles.

In [31], a frequency-domain approach was introduced, observing that deliberately applied forces tend to exhibit a lower frequency range than unintentional collision forces. This work was extended in [39] using the derivative of the spectral norm of external force/torque signals to refine contact distinction. More recently, [30] demonstrated the validity of contact detection between co-carried objects and external objects in a collaborative cable-suspended scenario. However,

the approach in [30] does not address human–cable collisions or their management. Similarly, earlier studies [31], [38], [39] rely on a single threshold for force measurements, rather than a more robust composite criterion. To the best of the authors’ knowledge, this is the first work to introduce a scoring-based method for contact distinction in CDPRS. It also systematically incorporates human–cable collision and corresponding management into the framework, thus broadening the scope of contact distinction for human–robot interaction.

B. Methodology

Based on the dynamic model in (2), the external wrench $\mathbf{w}_{eq}$ exerted on the MP can be estimated as:

$$\mathbf{W}_d \boldsymbol{\tau}_{mf} - \mathbb{I}_p \dot{\mathbf{v}}_d - \mathbf{C} \mathbf{v}_d + \mathbf{w}_{eq} + \mathbf{w}_g = 0, \quad (18)$$

where $\boldsymbol{\tau}_{mf}$ is the measured filtered cable tension, $\mathbf{W}_d$ is the desired wrench matrix, $\mathbb{I}_p$ is the platform’s spatial inertia, $\mathbf{C}$ is the matrix of Coriolis and centrifugal wrenches, and $\mathbf{w}_g$ is the gravitational wrench. The term $\mathbf{w}_{eq}$ is a rough estimate of the equivalent external wrench to be determined, i.e.,

$$\mathbf{w}_{eq} = [f^x \ f^y \ f^z \ m^x \ m^y \ m^z]^\top, \quad (19)$$

where f^i and m^i denote the forces and moments along each axis i , respectively. Note that for human–cable collisions, the dominant external effect is the force component. Although cable tensions can generate a moment about the geometric center of the MP, this moment is usually negligible compared to the forces.

1) *Frequency-Analysis*: To differentiate these scenarios more reliably, a frequency-based criterion is applied rather than relying solely on force thresholds [39]. At each time instant, a Fast Fourier Transform (FFT) is computed over a sliding window of N samples of the raw force signal $\mathbf{f}_{eq}$ (specifically, the first three elements of the estimated wrench $\mathbf{w}_{eq}$ in (18)), weighted by a half-Hann window w . Denoting f^j as the j -th component of $\mathbf{f}_{eq}$ (with $j = x, y, z$), its FFT is represented as $F^j(n)$, and the magnitude $F_\omega^j(m)$ of the discrete Fourier transform is computed as

$$F_\omega^j(m) = \left\| \sum_{n=0}^{N-1} w_n F^j(n) e^{-2\pi i \frac{m n}{N}} \right\|, \quad m = 0, \dots, N-1, \quad (20)$$

where the half-Hann window is defined as

$$w_n = \frac{1}{2} \left(1 - \cos \left(\frac{2\pi n}{2N-1} \right) \right), \quad n = 0, \dots, N-1. \quad (21)$$

This window function assigns higher importance to recent measurements while progressively de-emphasizing older data. The window length N is chosen based on the desired frequency resolution $\Delta\omega = \frac{1}{T_s N}$, where T_s is the sampling period. Frequencies below $\omega_{\min}$ are associated with slower, voluntary interactions, while frequencies above $\omega_{\max}$ are assumed to be sensor noise.

To capture meaningful collision-related dynamics, the analysis focuses on the frequency band $[\omega_{\min}, \omega_{\max}]$. The Spectral

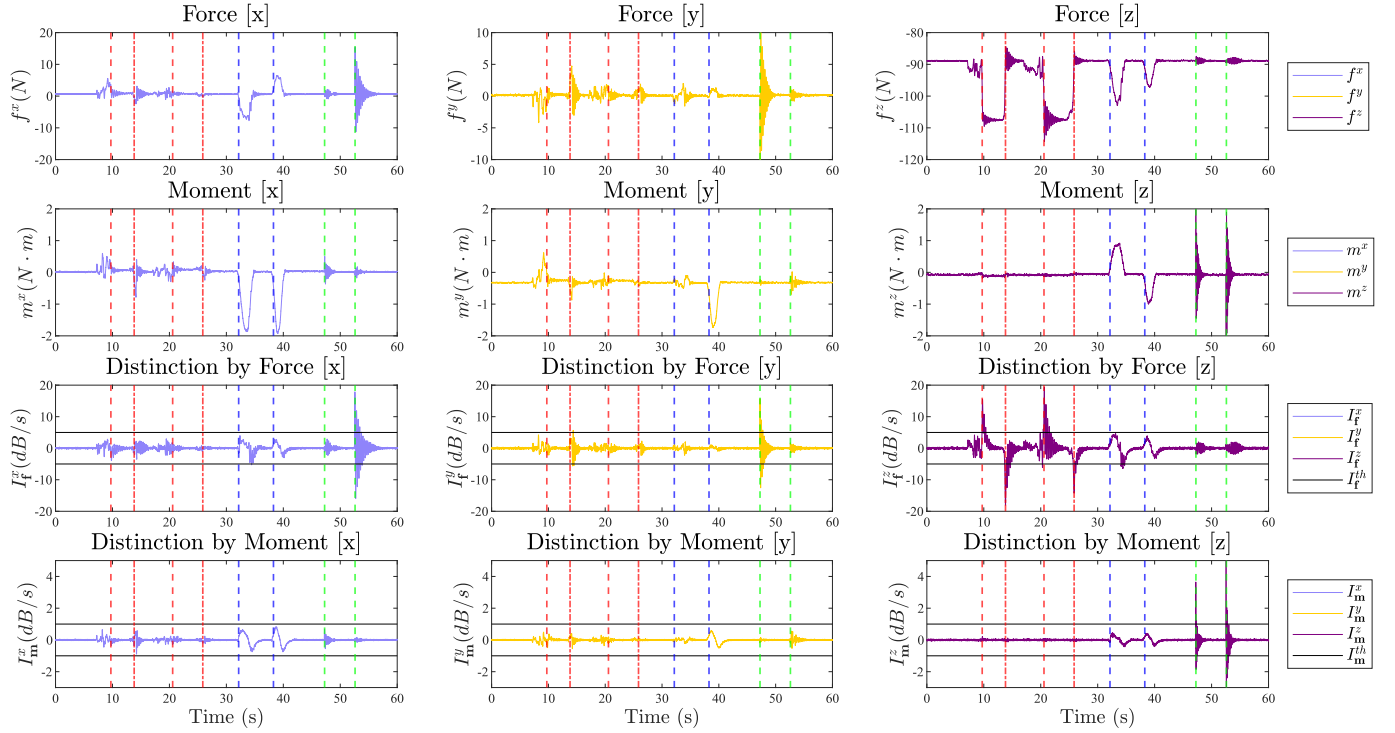


Fig. 6: Static Experiment: The top two rows display the estimated external forces and moments acting on the MP. The third and bottom rows show the spectral norm derivative (SND) indices for forces and moments along the x , y , and z axes. Colored vertical lines indicate detected contacts: red for SC1, blue for SC2, and green for SC3. The red dashed line indicates the payload placement, while the red dash-dotted line indicates the payload removal. Black lines in the third and fourth rows represent the proposed SND index thresholds for forces and moments, respectively.

Norm Derivative (SND) index for the j -th component, I_f^j , is defined as:

$$I_f^j = \frac{d}{dt} \left\| \mathbf{F}_{[\omega_{\min}, \omega_{\max}]}^j \right\|_1, \quad (22)$$

where $\mathbf{F}_{[\omega_{\min}, \omega_{\max}]}^j$ represents the sub-vector of frequency components within the chosen range, and $\|\cdot\|_1$ denotes the 1-norm (i.e., the sum of the magnitudes of these components). Hence, I_f^j captures the rate of change in the frequency content for the j -th axis. Similarly, I_m^j is obtained by applying the same procedure to the moment vector $\mathbf{m}_{\text{eq}} = [m^x, m^y, m^z]$ of the estimated wrench.

In this study, $\omega_{\min} = 3$ Hz is selected to exclude typical low-frequency motion of the MP (which is not related to collisions), and $\omega_{\max} = 50$ Hz is used to include human-robot interaction frequencies while omitting high-frequency noise. The sampling period is set to $T_s = 0.001$ s, and $N = 1024$ yields a window of 1.024 s. Thus, each SND index estimate leverages the most recent 1.024 s of wrench data, ensuring that significant transient events can be distinguished promptly.

2) *Distinction Criteria:* From the observations of SC1, SC2, and SC3, the SND indices for both force and moment components along the x , y , and z axes can be used to distinguish contact. Figure 6 illustrates the estimated external forces and corresponding SND indices for the static experiment described in Section IV. The figure is organized into four rows: the first row displays the force components, the second row shows the moment components, the third row presents the

SND indices for forces, and the fourth row presents the SND indices for moments. Each column corresponds to one of the Cartesian axes (x , y , z).

- **SC1 (Payload Placement/Removal):** When a payload is placed or removed, the force along the z -axis increases or decreases substantially, accompanied by minor transient oscillations. In most trials, in-plane components (x and y) remain low, although diagonal placement may induce more noticeable in-plane forces. Consequently, I_f^z tends to be larger, while $I_f^{x,y}$ are relatively small. Because the predominant force is parallel to the z -axis, I_m^z also remains small; similarly, $I_m^{x,y}$ are generally small when the payload is placed gently.
- **SC2 (Human–Cable Collision):** The forces evolve more gradually across all axes rather than showing abrupt changes. This behavior arises from the cable damping, which dissipates a portion of the external interaction force. Consequently, SC2 exhibits relatively small SND indices for both forces and moments.
- **SC3 (Human–MP Collision):** Transient oscillations appear in all axes when a collision occurs, although the amplitude in the z -axis is usually smaller, consistent with the assumption that vertical collisions are less frequent. Such interactions often yield larger SND indices for forces and moments along the x or y axis (or both). However, the amplitude distribution of these indices depends on the collision type and impact location; the maximum indices

may not always occur along the x or y axes exclusively. Additionally, an abrupt rotational interaction may produce a high I_m^z .

Based on these findings, Table I introduces a scoring-based framework for distinguishing contact scenarios. Each row corresponds to a scenario (SC1, SC2, SC3), and each column represents a condition checked against a chosen threshold. The table employs a “true” (✓) and “false” (✗) notation to indicate whether a specific force or moment criterion is exceeded. Weighted scores are assigned or subtracted based on the relevance of each condition: green cells indicate a score of 1 for high relevance, yellow cells represent a score of 0.5, and grey cells denote a score of 0.25 for partial relevance, depending on whether the condition is satisfied. After evaluating all conditions, the scenario with the highest cumulative score is declared the contact type. Furthermore, to ensure a fair comparison across scenarios, the cumulative scores are bounded between 0 and 1, where 1 corresponds to the highest relevance and 0 indicates no relevance. This method avoids ambiguity from overlapping criteria or the limitations of a single threshold. This is especially useful when multiple contact types exceed the threshold simultaneously, enhancing discrimination accuracy.

TABLE I: Scoring-Based Classification of Contact Events

Scenario	$I_f^{x,y} > I_f^{th}$	$I_f^z > I_f^{th}$	$I_m^{x,y} > I_m^{th}$	$I_m^z > I_m^{th}$
SC1	✗	✓	✗	✗
SC2	✗	✗	✗	✗
SC3	✓	✗	✓	✓

C. Results and Discussion

The threshold values for the SND indices are empirically determined: $I_f^{th} = 5 \text{ dB/s}$ for force and $I_m^{th} = 1 \text{ dB/s}$ for moment components. These thresholds are selected based on multiple contact experiments under varying tasks.

In the static experiment, by combining Fig. 5 and the SND index outcomes in Fig. 6, the following distinction results are obtained. Table II summarizes eight sequentially detected contact events. For instance, “Exp. j ” denotes the j^{th} contact event detected during the experiment; Exp. j^+ indicates a mass placement, while Exp. j^- corresponds to a mass removal. Each row lists the computed scores for SC1, SC2, and SC3, followed by the final distinction results. The results show that the predicted contact scenario matches the actual contact in each trial.

Additional experiments (80 in total) were conducted under dynamic conditions, including tests with varying payload masses, different force levels during human–cable collisions, and human–MP collisions from multiple directions. In every case, the proposed method reliably distinguished among SC1, SC2, and SC3, thereby demonstrating its robustness in real-world applications. The data from these experiments and the scripts are available as open source ¹.

TABLE II: Classification Results for Static Experiments

Experiments	Normalized scores			Results
	SC1	SC2	SC3	
Exp.1 ⁺ ✗✓✗✗	1.00	0.75	0	SC1
Exp.1 ⁻ ✓✓✗✗	0.90	0.5	0.18	SC1
Exp.2 ⁺ ✗✓✗✗	1.00	0.75	0	SC1
Exp.2 ⁻ ✗✓✗✗	1.00	0.75	0	SC1
Exp.3 ✗✗✗✗	0.60	1.00	0.36	SC2
Exp.4 ✗✗✗✗	0.60	1.00	0.36	SC2
Exp.5 ✓✗✗✓	0.10	0.50	0.81	SC3
Exp.6 ✓✗✗✓	0.10	0.50	0.81	SC3

D. Limitations and Future Work

Despite promising results, a few limitations remain. First, the experiments reported here involved controlled force levels and a limited number of subjects, partly due to safety restrictions when conducting pHRI. To address these constraints, future studies will expand participant diversity under formal ethical approval and enhanced safety protocols, allowing broader exploration of collision types and force ranges. In addition, optimizing the threshold and weighting coefficients in the scoring algorithm through data-driven or machine learning approaches may further improve discrimination accuracy. Such efforts will require larger datasets that capture diverse scenarios of human–robot collaborative transportation. Training can then be performed using numerical simulations that leverage a multi-physics simulator of the Cable-Driven Parallel Cobot (CDPC) to generate realistic data.

VI. CONTACT MANAGEMENT

In Subsection II-B, a workflow for managing contact scenarios during carrying tasks was introduced. Once a contact is detected and its scenario (SC1, SC2, or SC3) identified, specific actions are taken: payload mass estimation during placement (SC1), releasing the collided cable (SC2), or modifying the trajectory (SC3). While Section V presented static experimental results without the full management framework shown in Fig. 4, this section details the management processes and the criteria used to conclude each phase based on comprehensive experiments. In a 60-second dynamic trajectory, a 1.00 kg dumbbell is placed on the MP, followed by a human–cable collision, a human–MP collision, and finally, the dumbbell is removed. This experiment, which integrates all three scenarios—SC1 (payload placement and removal), SC2, and SC3—reflects real industrial and logistics applications and also safety concerns. Supplementary videos and accompanying data, demonstrating the full carrying task experiment, are available ².

A. SC1: Payload Placement/Removal

Referring to Fig. 4, the red blocks and text denote the additional components necessary for implementing these mod-

¹For further details and data access, please visit <https://uncloud.univ-nantes.fr/index.php/s/XJyc5tb4oDBCXzd>.

²These materials can be accessed at <https://uncloud.univ-nantes.fr/index.php/s/8d7oRdc9PCQ7ft2>.

ifications in the overall control scheme. In case a payload is placed on the MP without slipping, the overall suspended mass increases accordingly. If this increased mass is not accounted for, two primary issues arise:

- The IGEM becomes inaccurate, leading to elevated position tracking errors.
- The dynamic wrench in (10) is miscalculated, causing high tension tracking errors. These tension errors also serve as a general contact detection indicator; hence, unadjusted mass estimation would lead to persistent false contact signals.

To address these issues, the effective total mass of the MP and payload, denoted m_e , is estimated by modifying (2):

$$\mathbf{W}_d \boldsymbol{\tau}_{mf} - \mathbb{I}_p \dot{\mathbf{v}}_d - \mathbf{C} \mathbf{v}_d + \mathbf{w}_g^e = 0, \quad (23)$$

where $\mathbf{w}_g^e$ is the estimated gravitational wrench defined as

$$\mathbf{w}_g^e = \begin{bmatrix} m_e \mathbf{I}_3 \\ \hat{\mathbf{M}} \mathbf{S}_p^e \end{bmatrix} \mathbf{g}. \quad (24)$$

Here, $\mathbf{M} \mathbf{S}_p^e$ represents the new estimated first moment of inertia for the system. In practice, the MP rarely rotates about the x or y axes during carrying tasks, rendering the variations in the first moment of inertia negligible, i.e., $\mathbf{M} \mathbf{S}_p^e \approx \mathbf{M} \mathbf{S}_p$. This simplification allows us to estimate m_e by solving

$$\underset{m_e}{\operatorname{argmin}} \left\| \mathbf{W}_d \boldsymbol{\tau}_{mf} - \mathbb{I}_p \dot{\mathbf{v}}_d - \mathbf{C} \mathbf{v}_d + \mathbf{w}_g^e \right\|^2. \quad (25)$$

To prevent abrupt changes from immediately affecting the IGEM, an EMA filter with smoothing factor $\alpha = 0.01$ is applied:

$$m_{ef}(t) = \alpha m_{ef}(t) + (1 - \alpha) m_{ef}(t - 1). \quad (26)$$

Figure 7 illustrates an example in which a 1.00 kg payload is added to a 9.05 kg MP while it follows a reference trajectory. The blue curve indicates the actual total mass, and the red curve represents the filtered estimate. Although the estimation exhibits an initial overshoot, upon reaching steady state, the estimation error remains below 0.01 kg, demonstrating the effectiveness of this mass management approach.

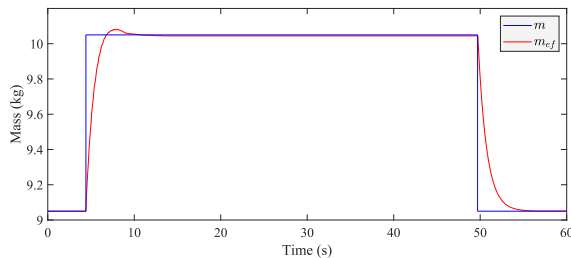


Fig. 7: Estimated filtered mass of the MP during payload placement and removal

1) *Management of Payload Placement/Removal:* Collisions involving the cable and the MP affect both $\mathbf{W}_d$ and $\boldsymbol{\tau}_{mf}$, which in turn compromise the accuracy of the mass estimation provided by (25). To mitigate this issue, mass estimation is performed only briefly upon detection of payload

placement/removal rather than continuously during the entire carrying task. Once SC1 is detected, the system monitors whether the effective mass m_{ef} remains within 0.005 kg for at least one second. When this stability criterion is met, the current estimate is maintained until the next SC1 is detected.

2) *Re-calibration of Payload Mass:* Upon activation of the special function, payload removal detection initiates a smooth exponential reversion of the estimated mass to the pre-measured MP mass.

B. SC2: Human–Cable Collision

Referring to Fig. 4, the blue blocks and text highlight the additional components required to manage the human-cable collision through release the collided cable. The method proposed in [27] is extended here, and the primary steps are summarized below:

- **Collision Detection and Cable Identification:** A collision time t_{col} is detected, and the specific cable c involved is identified [27]. The red detection block in Fig. 4 outputs $\boldsymbol{\tau}_{mf}$ and $\delta \boldsymbol{\tau}_f$ at t_{col} to identify the collided cable. A time function $s(t)$, with a 2-second transition, is then defined to avoid sudden reductions in tension. A selector matrix $\mathbf{SI}$ isolates the collided cable:

$$\mathbf{SI} = \operatorname{diag}(\delta_{1,c}, \delta_{2,c}, \dots, \delta_{8,c}), \quad (27)$$

where $\delta_{i,j} = 1$ if $i = j$, and 0 otherwise.

- **Gradual Reduction of the Collided Cable Tension:** Once the collided cable is identified, the proposed strategy in [28] reduces the desired cable tension using a tension distribution algorithm, blocks joint correction, and simultaneously increases the length of the collided cable to induce sagging. To ensure effectiveness, closed-loop cable tension tracking is employed. The management function $\mathbf{S}(t)$, obtained via $s(t) \cdot \mathbf{SI}$, decreases $\boldsymbol{\tau}_{\max}$ until reaching the minimum tension $\boldsymbol{\tau}_{\min}$. Concurrently, the cable release is calculated, transitioning the desired joint angle from $\hat{\mathbf{q}}_d$ to $\mathbf{q}_d$ by adding the optimal cable increment δl_c .
- **Detection of Collision End:** According to [28], the collision ends if δl_c remains below 0.1 mm for 1 second.
- **Post-Collision Restoration:** After detecting collision termination, $\mathbf{S}(t)$ is updated to gradually restore $\boldsymbol{\tau}_{\max}$ and return the cable length to its nominal value. This ensures a smooth transition back to normal operation, preventing discontinuities that may compromise reliability.

Figure 9 illustrates Cable 2 colliding with the human during dynamic motion in the WFW that allows tension release. Upon collision detection, the desired cable tension is decreased to 1 N before the collision ends. The tension profile indicates effective collision management, with tension reduction immediately following collision detection and restoration commencing one second later, fully recovering within two seconds. Throughout the collision, the tension of the involved cable remains under 5 N, a significant improvement compared to Fig. 5, where tension exceeded 40 N. This reduction enhances operator safety during human–robot interaction.

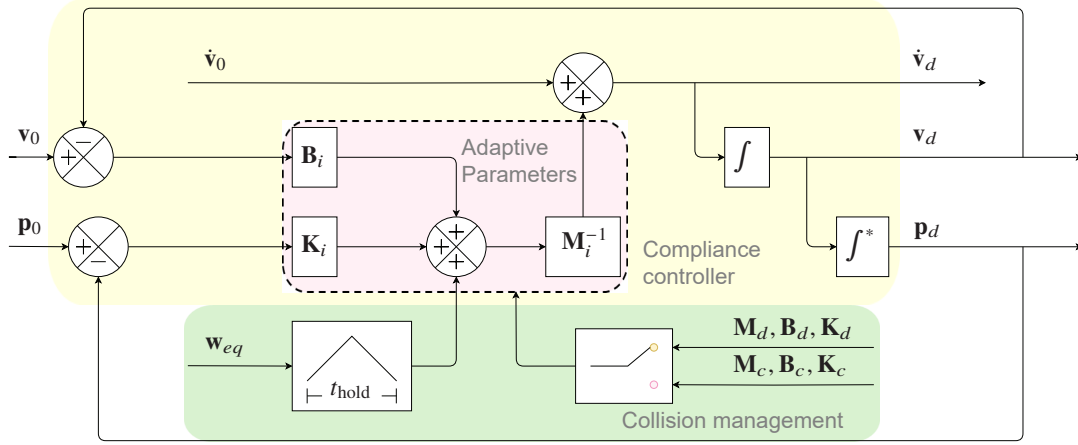


Fig. 8: Schematic of the adaptive compliant trajectory controller (detailed diagram as shown in Fig. 4). The yellow patch indicates the compliance controller operating under nominal conditions when no contact is detected. In the pink patch, the controller employs adaptive parameters based on the current contact scenario. The green patch represents the collision management, which selects the appropriate parameter set and incorporates a triangular buffer for the estimated external wrench w_{eq} along with the holding time t_{hold} to ensure a smooth transition back to the desired trajectory. Note that $\int$ denotes the integration symbol, and $\int^*$ represents the modified integration symbol that accounts for the rotation matrix.

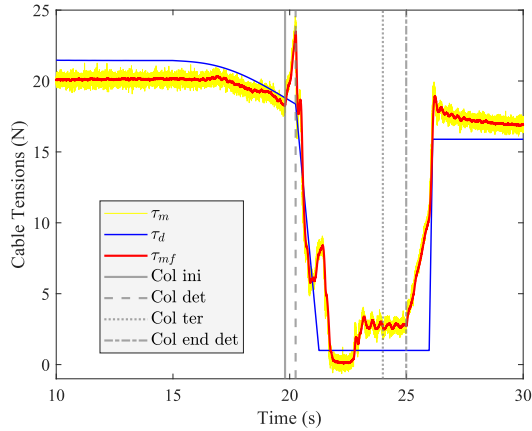


Fig. 9: Evolution of Cable 2 tensions and collision response under human-cable collision management. Here, ‘Col’ indicates collision, ‘ini’ indicates initialization, ‘det’ indicates detection, and ‘ter’ indicates termination.

C. SC3: Human–MP Collision

In real-world human–MP collision scenarios, the robot often lacks prior information regarding the human’s position, and visual data may be obscured. Consequently, the contact forces, either measured [16] or estimated using the collision identification algorithms shown in Fig. 4, can be employed to modify the desired trajectory in Cartesian space. Compliance control adjusts the desired trajectory in response to external forces acting on the cable. When a human–MP collision occurs, the general form of the compliance control equation [31] is given by

$$\mathbf{M}_i(\dot{\mathbf{v}}_0 - \dot{\mathbf{v}}_d) + \mathbf{B}_i(\mathbf{v}_0 - \mathbf{v}_d) + \mathbf{K}_i(\mathbf{p}_0 - \mathbf{p}_d) = \mathbf{w}, \quad (28)$$

where $\mathbf{M}_i$ denotes the inertia matrix, $\mathbf{B}_i$ the damping matrix, and $\mathbf{K}_i$ the desired stiffness matrix. The vectors $\mathbf{p}_d$, $\mathbf{v}_d$, and $\dot{\mathbf{v}}_d$ represent the desired position, velocity, and acceleration, respectively, while $\mathbf{p}_0$, $\mathbf{v}_0$, and $\dot{\mathbf{v}}_0$ denote the reference position, velocity, and acceleration. The term $\mathbf{w}$ represents the external wrench acting on the system.

Rearranging (28) to express the acceleration error yields

$$\dot{\mathbf{v}}_0 - \dot{\mathbf{v}}_d = \mathbf{M}_i^{-1} [\mathbf{w} - \mathbf{B}_i(\mathbf{v}_0 - \mathbf{v}_d) - \mathbf{K}_i(\mathbf{p}_0 - \mathbf{p}_d)]. \quad (29)$$

For a given net force discrepancy, a larger inertia matrix $\mathbf{M}_i$ reduces the magnitude of the acceleration error. However, increasing $\mathbf{M}_i$ also affects the dynamic response by lowering both the natural frequency and the damping ratio [31]. The natural frequency is defined as

$$\omega_n = \sqrt{\frac{\mathbf{K}_i}{\mathbf{M}_i}}, \quad (30)$$

and the damping ratio is defined as

$$\zeta = \frac{\mathbf{B}_i}{2\sqrt{\mathbf{K}_i\mathbf{M}_i}}. \quad (31)$$

This reduction results in a slower transient response. In industrial applications, high-precision acceleration tracking and rapid responsiveness to external forces are essential to move the MP away from the human operator when a collision occurs. Therefore, the adaptive compliant trajectory controller shown in Fig. 8 is employed.

In Fig. 8, the yellow patch indicates the compliance controller that is active when no contact is detected. In the pink patch, the compliant trajectory controller uses adaptive parameters depending on the contact scenarios. These parameters include the desired inertia $\mathbf{M}_d$, damping $\mathbf{B}_d$, and stiffness $\mathbf{K}_d$, while an alternative set of parameters, $\mathbf{M}_c$, $\mathbf{B}_c$, and $\mathbf{K}_c$, is applied when contact is detected and classified as a human–MP

collision. Accordingly, the following equations are obtained:

$$\mathbf{M}_d(\dot{\mathbf{v}}_0 - \dot{\mathbf{v}}_d) + \mathbf{B}_d(\mathbf{v}_0 - \mathbf{v}_d) + \mathbf{K}_d(\mathbf{p}_0 - \mathbf{p}_d) = 0, \quad (32)$$

$$\mathbf{M}_c(\dot{\mathbf{v}}_0 - \dot{\mathbf{v}}_d) + \mathbf{B}_c(\mathbf{v}_0 - \mathbf{v}_d) + \mathbf{K}_c(\mathbf{p}_0 - \mathbf{p}_d) = \mathbf{w}_{eq}, \quad (33)$$

where $\mathbf{w}_{eq}$ denotes the estimated external wrench acting on the system, obtained from (18). Equations (32) and (33) describe the system behavior for the cases without and with active collision management, respectively. The management phase terminates when the Cartesian distance between the reference and compliant trajectories remains below 0.01 m for 1 s. Note that purely orientational compliant trajectories are not considered in this study, as they do not enable the MP to move away from the human. However, such cases can be detected through cable tension error analysis, and additional functions can be implemented to address them.

The green patch in Fig. 8 represents the SC3 management for the compliance controller, where the parameter t_{hold} and the corresponding block constitute a triangular buffer for the estimated external wrench $\mathbf{w}_{eq}$. The rationale for this buffer is based on two considerations. First, although the adaptive compliance controller is primarily designed for human–robot collaboration [40], the management strategy employed herein aims to move the MP away from the human to prevent potential subsequent collisions. During compliant control, contact between the human and MP may cease and subsequently resume, thereby necessitating a holding mechanism. Second, when returning to the desired trajectory, the estimation of $\mathbf{w}_{eq}$ may exhibit significant errors; incorporating this noisy signal into (33) can result in inconsistent solutions and impulsive responses that may exceed the system’s control torque capacity or impose an excessive computational load, thereby destabilizing the MP.

In practice, the compliant parameters are selected based on the system properties; notably, the desired inertia $\mathbf{M}_d$ is taken as the measured inertia of the system. Accordingly, the collision inertia matrix is defined as

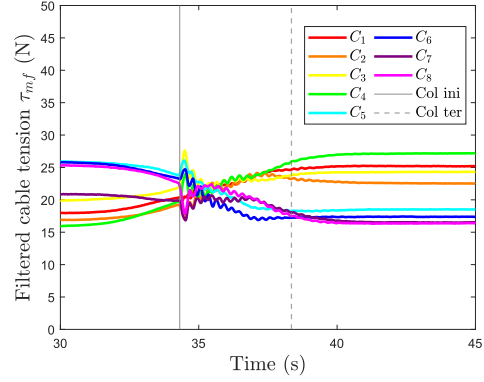
$$\mathbf{M}_c = \kappa \mathbf{M}_d, \quad (34)$$

with a small proportionality constant κ ensures that the MP moves away from the human effectively while avoiding persistent collisions. The damping and stiffness in the collision mode are defined as

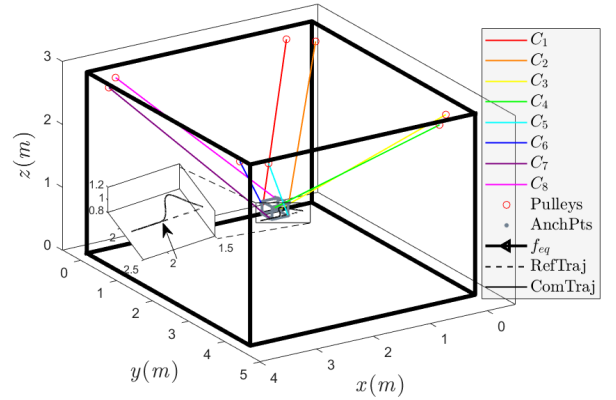
$$\mathbf{B}_c = \frac{\mathbf{B}_d}{\kappa}, \quad \mathbf{K}_c = \frac{\mathbf{K}_d}{\kappa}. \quad (35)$$

This configuration increases the natural frequency ω_n and the damping ratio ζ relative to their nominal values scaled by $\frac{1}{\kappa}$. The introduction of κ and t_{hold} provides a compromise that enables safe trajectory modification during collision events while preserving overall system stability.

Figure 10 illustrates the filtered cable tensions and the evolution of the reference trajectory (dashed line) alongside the compliant trajectory (solid black line). Upon detecting a human–MP collision, the MP smoothly adapts its path and eventually returns to the reference trajectory; the parameters $t_{\text{hold}} = 4$ s and $\kappa = 0.1$ ensure a safe deviation. In Fig. 10a, a slight oscillation is observed during the 4 s interval in



(a) Filtered cable tensions



(b) Reference and compliant trajectory

Fig. 10: Filtered cable tensions (top) and comparison of reference with compliant trajectory in space view (bottom).

which $\mathbf{w}_{eq}$ is applied, but the compliant trajectory gradually converges with the reference. The proposed method effectively directs the MP away from the operator by modifying the trajectory and then automatically resumes the nominal path once contact ends.

Future work will focus on optimizing the external wrench estimation and the computational efficiency of the compliance controller. An adaptive compliance controller that relies on real-time estimation is planned. This controller will not only adjust the Cartesian trajectory but also adapt the cable tension distribution to maintain MP stability, thus aligning with research on human–MP collaborative manipulation.

VII. CONCLUSION

This paper presented a comprehensive framework for detecting, distinguishing, and managing contact events in human-assisted CDPR carrying tasks. The method classifies three frequently encountered scenarios in collaborative carrying tasks: payload placement and removal, human–cable collisions, and human–MP collisions. Unlike traditional collision detection and distinction methods that rely on single force thresholds, the proposed approach employs a frequency-domain analysis of the estimated external wrench to compute the Spectral

Norm Derivative (SND) indices for force and moment components. These indices, combined with a scoring-based criterion, robustly distinguish the contact scenarios in real time, also under dynamic conditions. Experimental results showed that updating the robot's mass estimate when a payload is placed significantly improves trajectory tracking, while releasing tension on the collided cable enhances operator safety. Furthermore, a compliant controller effectively resolved human-MP collisions, allowing the robot to deviate from its planned path and safely return once the collision subsided.

From the experimental results, it is evident that the proposed method achieves high levels of safety and reliability without unduly compromising overall performance or productivity. However, several avenues remain open for future investigation. Although the present study treats human-cable and human-MP collisions strictly as safety threats, future work might reframe these interactions as opportunities for collaboration. This could be accomplished by leveraging the contact distinction method, along with refined threshold settings and scoring weight configurations. In addition, although preliminary evaluations validate the proposed approach, additional experiments should be conducted involving a broader group of participants and more diverse collision types. Formal ethical approval, in conjunction with enhanced safety protocols, is required to create realistic scenarios that emulate industrial settings. Data obtained from these expanded experiments can be analyzed to enhance transparency and validate the reliability of human-robot collaboration and safety [41]. In addition, data gathered from physical simulators can be leveraged within data-driven or machine learning frameworks to refine the thresholds and weighted scoring used in contact distinction, as well as to optimize the parameters of the compliant controller for improved adaptability.

Another assumption that warrants further exploration involves releasing cable tension in the WFW. When a cable collision occurs, the current method assumes that the collided cable can be safely released within its feasible range. Future studies will incorporate a trajectory-compliant controller that repositions the MP into a region of the workspace where cable release is possible if it is not initially feasible. This enhancement would bolster the versatility of the proposed architecture, making it applicable to a wider variety of industrial and logistic tasks.

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